

PROJECT

5.3

**CREATING A
PRIVATE SUBNET**
NETWORKING SERIES

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Objective

This project aims to create a private subnet in an AWS VPC to securely host internal resources without direct internet access. It ensures isolation and is ideal for sensitive services like databases or internal APIs.

To support this, a private route table is set up for internal routing, and a custom network ACL is configured to tightly control traffic flow. Together, these components enhance security and restrict unauthorized access.

Services

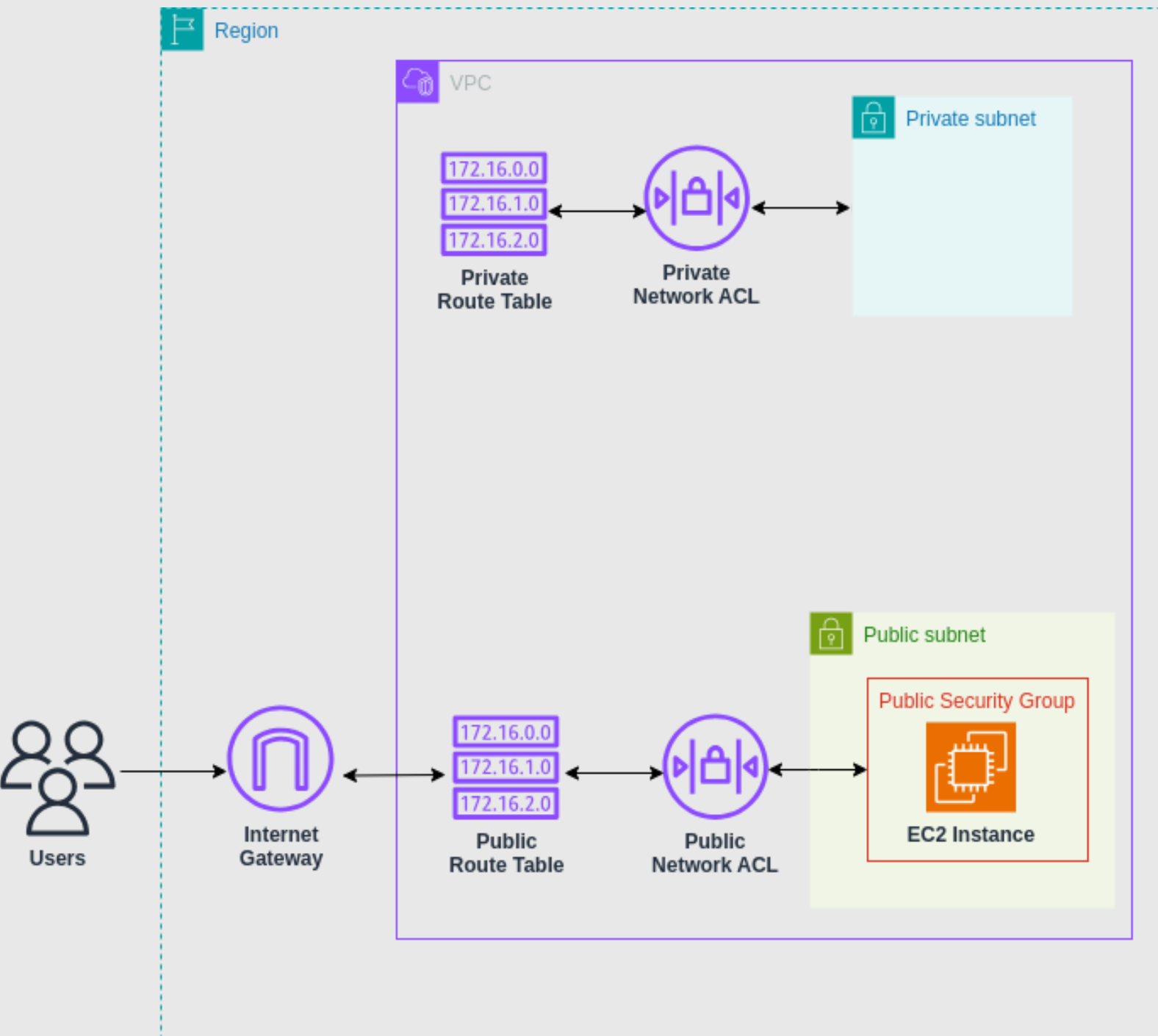
Amazon VPC - Enables creation of a logically isolated virtual network in the AWS cloud.

Subnets - Divide the VPC into smaller, manageable sections for organizing resources.

Route Tables - Control the flow of traffic within the VPC and between subnets.

Network ACLs - Provide stateless traffic filtering at the subnet level for added security.

Architecture Diagram



The architecture includes a VPC with public and private subnets. The private subnet is linked to a private route table for internal only routing. A custom network ACL adds an extra layer of security by controlling traffic flow.

Steps to Implement

1. Create a VPC:

Start by creating your own Virtual Private Cloud (VPC), which is like setting up your own private network inside AWS. You can define the IP address range using CIDR blocks (like 10.0.0.0/16). This step gives you full control over your cloud networking environment like a secure container for all your AWS resources.

2. Create a Private Subnet:

Within the VPC, create a private subnet using a smaller CIDR block (like 10.0.1.0/24). Ensure the subnet has no public IP auto assignment. This subnet is used to host resources that should not be directly accessible from the internet.

3. Create a Private Route Table:

Set up a custom route table and associate it only with the private subnet. Do not add any internet gateway routes. This ensures traffic stays internal, either within the VPC or directed to specific AWS services via NAT or endpoints if needed.

4. Create a Private Network ACL:

Create a custom network ACL and attach it to the private subnet. Define specific inbound and outbound rules to control traffic at the subnet level. This provides stateless, layer level security to further isolate and protect your resources.

Story Time

VPC (Your City Walls):

This is like building strong walls around your city. You decide how big your city is by choosing the area (IP range). Only invited guests can enter, making it your own secure kingdom in the cloud!

Private Subnet (Secret Neighborhood):

Think of this as a hidden, VIP neighborhood inside your city where important people (like databases) live. There are no direct roads to the outside world, it's peaceful, private, and protected.

Private Route Table (Local Traffic Map):

This is your neighborhood's custom map. It tells vehicles (data) where they're allowed to go only within the city! No highways to the internet here, just internal routes for safe travel.

Network ACL (Security Guards at the Gate):

These are your trusty guards who check every vehicle at the gate. They follow strict rules on who can enter or leave, ensuring only safe, approved traffic moves in or out of the VIP area.

BY
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THANK YOU!

